



Figure 1:

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1 "ctype" library

The ctype library in C is used for character classification. It provides functions to categorize characters into different types, such as alphabetic characters, digits, punctuation characters, whitespace characters, and control characters.

most commonly used functions in the ctype.h library:

isalpha(c): checks if the character c is an alphabetic character (either lowercase or uppercase).

isdigit(c): checks if the character c is a digit (0-9).

islower(c): checks if the character c is a lowercase letter.

isupper(c): checks if the character c is an uppercase letter.

isalnum(c): checks if the character c is either an alphabetic character or a digit.

isxdigit(c): checks if the character c is a hexadecimal digit (0-9, a-f, A-F).

isspace(c): checks if the character c is a whitespace character (space, tab, newline, etc.).

isctrl(c): checks if the character c is a control character (non-printable character).

isgraph(c): checks if the character c is a printable character, excluding space.

isprint(c): checks if the character c is a printable character, including space.

tolower(c): converts the character c to lowercase.

toupper(c): converts the character c to uppercase.

These functions are very useful for validating user input, parsing strings, and performing other common text processing tasks in C.

2 "stdlib" / standard library

stdlib, also known as standard library contains following functions

malloc(size): dynamically allocates a block of memory of the specified size and returns a pointer to the beginning of the block.

calloc(nmemb, size): dynamically allocates a block of memory for an array of nmemb elements, each of which has a size of size bytes, and initializes the memory to zero.

realloc(ptr, size): changes the size of the previously allocated block of memory pointed to by ptr to the specified size.

free(ptr): deallocates the block of memory pointed to by ptr, which was previously allocated by malloc(), calloc(), or realloc().

rand(): generates a random number.

srand(seed): initializes the random number generator with the specified seed value.

exit(status): terminates the program with the specified exit status.

atexit(func): registers the function func to be called at normal program termination.

system(command): executes the specified shell command.
abs(value): returns the absolute value of the integer value.
labs(value): returns the absolute value of the long integer value.
div(numer, denom): performs integer division of the numerator numer by the denominator denom and returns a structure containing the quotient and remainder.
ldiv(numer, denom): performs long integer division of the numerator numer by the denominator denom and returns a structure containing the quotient and remainder.

3 conversion functions

atoi(str): converts a string str representing an integer to an integer value.
atof(str): converts a string str representing a floating-point number to a double value.
atol(str): converts a string str representing a long integer to a long integer value.
strtol(str, endptr, base): converts a string str representing an integer in the specified base (2-36) to a long integer value. The endptr argument is a pointer to a variable that will receive the address of the first character in str that could not be converted to an integer.
strtoul(str, endptr, base): converts a string str representing an unsigned integer in the specified base (2-36) to an unsigned long integer value. The endptr argument is a pointer to a variable that will receive the address of the first character in str that could not be converted to an unsigned integer.
sprintf(str, format, ...): formats the output according to the format specification in format and stores it in the string str. The ... argument is used to pass additional arguments to the function.
snprintf(str, n, format, ...): same as sprintf(), but limits the number of characters written to str to n.
sscanf(str, format, ...): reads formatted input from the string str according to the format specification in format and stores it in the specified variables. The ... argument is used to pass additional arguments to the function.

4 "string" functions

strcmp(str1, str2): Compares two strings and returns 0 if they are equal, a positive value if str1 is greater than str2, and a negative value if str1 is less than str2.
strncmp(str1, str2, n): Compares the first n characters of two strings and returns 0 if they are equal, a positive value if str1 is greater than str2, and a negative value if str1 is less than str2.
strlen(str): Returns the length of a string (excluding the null character).

strcpy(dest, src): Copies the contents of src to dest.
strncpy(dest, src, n): copies the first n characters of src to dest.
strcat(str1, str2): Concatenates str2 to the end of str1.
strncat(str1, str2, n): Concatenates the first n characters of str2 to the end of str1.
strstr(str1, str2): Returns a pointer to the first occurrence of str2 in str1, or NULL if str2 is not found.
strchr(str, c): Returns a pointer to the first occurrence of character c in str, or NULL if c is not found.
strrchr(str, c): Returns a pointer to the last occurrence of character c in str, or NULL if c is not found.
strtok(str, delim): Splits a string into tokens based on a delimiter and returns a pointer to the first token.
strupr(str): Converts a string to uppercase.
strlwr(str): Converts a string to lowercase.

5 Memory Function

memory management functions are essential for dynamically allocating and deallocating memory in C programs. Proper use of these functions helps prevent memory leaks and ensures efficient memory usage.

strcmp(str1, str2): Compares two strings and returns 0 if they are equal, a positive value if str1 is greater than str2, and a negative value if str1 is less than str2.

strncmp(str1, str2, n): Compares the first n characters of two strings and returns 0 if they are equal, a positive value if str1 is greater than str2, and a negative value if str1 is less than str2.

strlen(str): Returns the length of a string (excluding the null character).

strcpy(dest, src): Copies the contents of src to dest.

strncpy(dest, src, n): Copies the first n characters of src to dest.

strcat(str1, str2): Concatenates str2 to the end of str1.

strncat(str1, str2, n): Concatenates the first n characters of str2 to the end of str1.

strstr(str1, str2): Returns a pointer to the first occurrence of str2 in str1, or NULL if str2 is not found.

strchr(str, c): Returns a pointer to the first occurrence of character c in str, or NULL if c is not found.

strrchr(str, c): Returns a pointer to the last occurrence of character c in str, or NULL if c is not found.

strtok(str, delim): Splits a string into tokens based on a delimiter and returns a pointer to the first token.

strupr(str): Converts a string to uppercase.

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6 c programming input and output

printf() function is used to display output on the console. It can display various data types like integers, floats, characters, and strings. It uses format specifiers to display the output.

scanf() function is used to read user input from the keyboard. It can read values of various data types like integers, floats,